

Chun Song

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Education

The Hong Kong University of Science and Technology (Guangzhou) FULLY FUNDED PH.D. STUDENT, URBAN GOVERNANCE AND DESIGN (UGOD)	<i>Guangzhou, P.R. China</i> <i>Sep. 2025 - Present</i>
Institute of Engineering Mechanics, China Earthquake Administration MASTER OF ENGINEERING IN STRUCTURAL ENGINEERING	<i>Harbin, P.R. China</i> <i>2022 - 2025</i>
Beijing Jiaotong University B.E. IN CIVIL ENGINEERING, SCHOOL OF CIVIL ARCHITECTURE AND ENGINEERING	<i>Beijing, P.R. China</i> <i>2018 - 2022</i>

Research Interests

Core Research Interests: Multi-agent systems, disaster simulation, and crowd dynamics.

Core Research Question: How can we develop generalizable, verifiable simulations and effectively integrate AI to enable prediction, optimization, and decision-making?

Research Affiliations

Responsible Urban Intelligence Lab (RUI Lab), HKUST(GZ) RESEARCH ASSISTANT, SUPERVISED BY PROF. RUI CAO	<i>Guangzhou, P.R. China</i> <i>Jul. 2025 - Aug. 2025</i>
Chongqing University Liyang Smart City Research Institute RESEARCH STUDY	<i>Liyang, P.R. China</i> <i>2023 - 2025</i>
Harbin Institute of Technology, School of Civil Engineering RESEARCH STUDY	<i>Harbin, P.R. China</i> <i>2022 - 2023</i>

Research Funding

Guangdong Provincial Graduate Student Research Project	2026
• Project Lead; RMB 20,000 awarded for “Physics-Simulation-Driven Digital-Twin Scenario Deduction and Emergency Decision-Making Methods for Community Indoor Fires.”	

Honors and Awards (Selected)

2026	Third Prize (two projects) , Urban Cup 2026. Award List “Governing the AI Shock: From Laissez-Faire Disruption to a Broad-Based AI Social Compact.” Project Poster Competition Video “Personal Shade, Shared Consequences: How Umbrellas Reshape Movement and Solar Exposure in Dense Public Spaces.” Project Poster.
2025	Excellent Simulation Technology Solution , 2025 Typical Case Collection, for “Large-scale Multi-hazard Digital Simulation and Scenario-deduction System for Cities and Urban Agglomerations.” Certificate Project Poster .
2025	Excellent Presentation Award , the 7th National Conference on Disaster Prevention and Mitigation Engineering, for “Urban Emergency Evacuation Simulation Models and Systems.” Conference Program Certificate .
2024	Second Prize , Postgraduate Division, the 4th Engineering Simulation Innovation Design Competition, 2024 Belt and Road and BRICS Skills Development and Technology Innovation Competition. Certificate .
2023, 2024	First-class Graduate Academic Scholarship , Institute of Engineering Mechanics (two-time recipient).
2022	Second-class Graduate Academic Scholarship , Institute of Engineering Mechanics.
2021	Honorable Mention , 2021 Interdisciplinary Contest in Modeling (ICM). Team: Yingying Han, Yuqin Zhou, and Chun Song; faculty advisor: Guozhong Liu. Certificate .

Publications ([†] corresponding author)

[P5] Coupled Density-Hazard Navigation for Evacuation Reliability in Dense Urban Informal Settlements. [RESS Poster](#)

Chun Song, Xuchuan Lin, Rui Cao[†].

Accepted for publication in Reliability Engineering & System Safety; forthcoming online.

[P4] A Simulation-Based Framework for Optimizing Road Networks for Disaster Evacuation in Dense Informal Urban Settlements. [CEUS Poster](#)

Chun Song, Zhijie Zhou, Rui Cao[†].

Under review at Computers, Environment and Urban Systems (CEUS).

[P3] Unveiling Inequalities in Forgotten Enclaves: A Geospatial-Streetscape Integrated Framework for Assessing Urban Village Livability.

Zhijie Zhou, Chun Song, Rui Cao[†].

Under review at Cities.

[P2] Analysis of influencing parameters of building fire spread simulation in urban area based on cellular automata.

Ning Yang, Qi Wang, Jiangrong Zhong, Xuchuan Lin[†], Chun Song.

The 18th China CAE Engineering Analysis Technology Annual Meeting and 14th China Digital Simulation Forum, in Chinese.

[P1] Limited Monitoring Data Based Updating Method for Fundamental Period of the Building Group Model.

Chen Xianan, Lin Xuchuan[†], Zhang Lingxin, Gao Wuping, Song Chun.

Engineering Mechanics, 2024, in Chinese.

Conference Presentations ([†] corresponding author)

[Poster 2] Coupled Density-Hazard Navigation for Evacuation Reliability in Dense Urban Informal Settlements.

Chun Song, Xuchuan Lin, Rui Cao[†].

Poster accepted at the 4th International Conference on Urban Science and Intelligence, Guangzhou, China, August 8-9, 2026. [Conference](#) | [RESS Poster](#).

[Poster 1] A Simulation-Based Framework for Optimizing Road Networks for Disaster Evacuation in Dense Informal Urban Settlements.

Chun Song, Zhijie Zhou, Rui Cao[†].

Poster accepted at the 6th CAAI Symposium on Social Computing and Social Intelligence (SCII Summit 2026), Guangzhou, China, April 17-18, 2026. [Conference Report](#) | [CEUS Poster](#).

[Oral 1] Modeling and Simulation of Urban Crowd Emergency Evacuation.

Chun Song, Yue Wang, Xuchuan Lin[†].

Oral presentation at the 3rd Academic Conference on Digital Twin and Intelligent Construction, December 15, 2024. [Presentation Certificate](#).

Intellectual Property

[Patent 2] Road-widening optimization method, system, and storage medium driven by emergency-evacuation simulation diagnosis.

Rui Cao, Chun Song, Zhijie Zhou.

Chinese invention patent, granted June 9, 2026. Patent No. ZL 2026 1 0338721.6. [Patent Certificate](#).

[Patent 1] Simulation method, system and computer storage medium of earthquake secondary fire spread.

Xuchuan Lin, Qi Wang, Jiangrong Zhong, Chun Song, Ning Yang.

Published.

[Software Copyright 1] Crowd Emergency Evacuation Simulation Software V1.0.

Chun Song, Xuchuan Lin, Jiangrong Zhong, Qi Wang, Xianan Chen, Xiaoxiang Yuan, Xiaonan Zhang.

Computer software copyright, registered November 19, 2024. Registration No. 2024SR1825682. [Certificate](#).

Skills

Programming Python, C++.

Research Physics-based simulation modeling, GIS, reinforcement learning.

Academic Services

- 2026-27 Fall Graduate Teaching Assistant, *Introduction to GeoAI* (UGOD 5300), The Hong Kong University of Science and Technology (Guangzhou).
- 2025-26 Spring Teaching Assistant for Undergraduate Course, *History of Chinese Civilization: From Antiquity to Late Imperial* (UCUG 1604), The Hong Kong University of Science and Technology (Guangzhou).